PROJECT: HOSTFALL

Game Design & Build Specification — Claude Code Handoff

A retro single-screen sci-fi action game in the spirit of Berzerk and the original Legend of Zelda. Four acts, ~15-20 minutes per playthrough, browser-based.

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1. CORE VISION

You're a soldier sent into a compromised military base. Surface communications cut, every recon team lost. You fight in, discover the base hosts an alien presence that's gone deeper than anyone realized, and end up sprinting out as the place collapses.

Tone: Tense, fast, escalating. Starts gritty-military, ends body-horror sci-fi. Quick deaths, retry-friendly, but with a real arc.

Reference touchstones: Berzerk (room-by-room, hostile-everywhere), Smash TV (combat density), early Zelda (screen transitions, weapon progression), the movie *Aliens* (recon → discovery → descent → escape pacing).

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2. TECH STACK — COMMITTED CHOICES

Build target: Single self-contained `index.html` file with embedded `<style>` and `<script>` tags. No external dependencies. No build step. No frameworks. Open the file in a browser, the game runs.

Rendering: HTML5 Canvas 2D. Internal resolution 320×240, scaled up via CSS to fit the viewport with `image-rendering: pixelated`. Use a separate offscreen canvas at native resolution and blit to a display canvas to keep integer scaling clean.

Sprites: No external image files. All sprites drawn via canvas primitives using small color-data arrays (2D arrays of palette indices). Helper function `drawSprite(spriteData, palette, x, y, scale)` paints them as small rects. Player ~12×16 px. Enemies ~12×16 to 24×24. Bosses ~32×32 to 48×48.

Audio: Web Audio API, all procedural. Oscillators + noise generator + envelope shapes. No audio files. Per-act ambient drone (low sine + filtered noise) instead of music.

Input: Keyboard only. WASD movement. Arrow keys aim/shoot direction (twin-stick on keyboard). Space throws grenade. Shift sprints. E interacts/picks up. Tab cycles weapon. R reloads. Escape pauses.

Persistence: None. Death = restart current act. Act completion = in-memory checkpoint only. No localStorage.

3. SCREEN ARCHITECTURE

The world is a graph of single-screen rooms. Each room is exactly one viewport — no scrolling.

Room object:

```js
{
  id: "act1_r03",
  act: 1,
  walls: [{x, y, w, h}, ...],          // obstacle rects
  exits: { n: "act1_r04", s: null, e: "act1_r02", w: null },
  enemies: [{type, x, y}, ...],         // spawn list
  pickups: [{type, x, y}, ...],
  cleared: false,
  type: "combat" | "puzzle" | "story" | "transition" | "boss",
  lockExitsUntilCleared: true,
  bgColor: "#1a1a1a",
  wallStyle: "concrete" | "metal" | "organic" | "alien"
}
```

Transitions: Walk off an edge with an exit → instant cut with 100ms black flash → spawn 16px in from the opposite edge of the new room.

Locked exits: Combat rooms lock exits until cleared. Visual: red barrier across doorways, turns open when cleared. Plays a "clear" sound (rising blip).

Room counts:

- Act 1: 10 screens (8 combat/exploration, 1 story beat, 1 boss)
- Act 2: 10 screens (7 combat, 2 environmental/puzzle, 1 boss)
- Act 3: 8 screens (6 hostile, 1 hazard gauntlet, 1 boss)
- Act 4: 8 screens (escape — warped earlier rooms with wave spawners)

Total: 36 screens.

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4. PLAYER MECHANICS

Health: 5 HP. Damage is binary — 1 HP per hit regardless of source. Med-stim pickups restore 2 HP; placed every 4-5 screens.

Invulnerability frames: 60 frames (1s at 60fps) after taking damage. Player sprite flashes during i-frames.

Movement: 8-directional, free (not tile-locked). Base speed 90 px/s. Sprint = 1.5× for 2s before stamina depletes; regen 4s when not sprinting.

Shooting: Aim independent of movement (twin-stick). Press an arrow key = face that way and fire on the configured rate. Holding the key = continuous fire.

Collision: Player hitbox 8×8 centered on the sprite. Walls block. Enemies don't push but contact damages.

Death: Sprite flashes red, brief particle burst, fade to black, "RESTARTING ACT" for 1.5s, respawn at act start. Unlimited respawns.

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5. WEAPONS — DEFINITIVE STATS

Weapons stack. Tab cycles through owned weapons. Found weapons stay in inventory across deaths within the same run.

#	Weapon	Damage	Fire rate	Ammo	First found	Notes
1	Pulse Rifle	1	4/s	∞	start	Default workhorse
2	Pistol	1	6/s	∞	act 1 r2	Faster fire, slightly slower bullets
3	Plasma Rifle	2	2/s	30 (max 60)	act 2 r1	Pierces 1 enemy, slower projectile
4	Heavy Laser	4	1/s	8 (max 16)	act 3 r2	Instant beam, knockback, boss-killer
G	Grenade	3 AoE	manual	3 (max 5)	start (1)	Space button. 0.8s fuse, 32px blast

Projectile speeds: pulse 240 px/s, pistol 200 px/s, plasma 160 px/s. Slower projectiles trade off for higher damage. Grenade arcs in aim direction with 0.8s fuse and 32px blast radius (full damage in inner 16px, half outside that).

Ammo pickups: small (5 plasma / 2 laser), large (15 plasma / 4 laser). Roughly one ammo pickup per 2-3 screens once the relevant weapon is acquired.

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6. ENEMY ROSTER

Total: 10 enemy types + 4 bosses. Each act introduces 2-3 new types.

Act 1 — Hostile Soldiers

- **Grunt** (12×16, khaki). 2 HP. Speed 50 px/s. Patrols, line-of-sight detect, fires single bullet at 120 px/s for 1 dmg. Stops to fire. Berzerk-style baseline.
- **Sergeant** (14×18, dark fatigues + red beret). 3 HP. Speed 60 px/s. Burst-fires 3 shots with 0.4s pause. Uses walls for cover (moves toward nearest wall edge after firing).
- **Twitchy** (12×16, soldier with visible bulges). 2 HP. Speed 110 px/s. Doesn't shoot — charges player and contact-damages. Foreshadows act 2 hybrids.

Act 2 — Hybrids & Aliens

- **Hybrid Soldier** (14×18, ragged uniform with carapace). 4 HP. Speed 70 px/s. Fires 2-shot bursts, occasionally lunges if within 60px.
- **Alien Grunt** (14×16, chitinous, no clothes). 3 HP. Speed 100 px/s. Pure melee charger, faster than Twitchy and harder to kill.
- **Spore Drone** (12×12, bloated/round). 1 HP. Speed 30 px/s. Slow approach, explodes on death or contact, releasing a 24px-radius spore cloud that lingers 3s and damages player on contact.

Act 3 — Pure Alien

- **Stalker** (14×16, sleek/insectoid). 3 HP. Speed 130 px/s. Fast charger that strafes around player at close range. Hard to hit.
- **Spitter** (16×16, anchored organism). 4 HP. Speed 0 (stationary). Spits arcing acid every 1.2s; acid leaves 2s damage puddle on the ground.
- **Drifter** (12×12, floating). 2 HP. Floats through some walls (specifically marked "porous" walls). Fires homing slow projectile (90 px/s, 1 dmg, slight tracking).

Act 4 — Escape Wave Units

- **Swarmer** (8×10, small alien). 1 HP. Speed 140 px/s. Spawned in waves of 4-8 from spawn points. Pure melee.
- (Plus all act 2 and 3 enemies as wave units, in mixed compositions.)

AI Behavior Notes

All shooting enemies should occasionally miss on purpose — fire toward a position 8-16 pixels off the player. Otherwise the game is unfair.

Pathing: simple — direct line toward player, break around walls by sliding along them. No A*. Berzerk-tier dumb-but-deadly is the target.

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7. BOSSES — FULL SPECIFICATIONS

Act 1 Boss: General Hayes / The Vessel

Arena: Command center. Larger room (clears two screen widths if you can; otherwise one screen with no internal walls, just desks/cover at the corners).

Phase 1 — General Hayes (human form): 8 HP. Sprite 16×20, officer in dress uniform with sidearm. Speed 70 px/s. Fires 3-shot bursts, takes cover behind desks. Talks during fight via on-screen dialogue text: "You shouldn't have come here." / "We're past saving." / "You're already too late."

Transition: At 0 HP, screen freezes for 1s, Hayes sprite convulses (flash red/white), explodes outward. Replaced by The Vessel sprite. Phase 2 starts with full new health bar.

Phase 2 — The Vessel (alien controller bursting from human): 12 HP. Sprite 24×24, chitinous with vestigial human features. Speed 90 px/s. Two attacks: (a) charge the player at 140 px/s for 1.5s if within 80 px, (b) ranged spit (120 px/s projectile, 1 dmg) every 1.5s. Drops a key card that opens the elevator.

Act 2 Boss: Brood Mother

Arena: Wet biological chamber. Three node positions: top-left, top-right, bottom-center. Brood Mother in middle (stationary).

Mechanic: Brood Mother is invulnerable while ANY of 3 pulsing nodes are alive. Kill all nodes → 10-second window where Brood Mother takes damage. Then nodes respawn, repeat. Three node-cycles to kill her.

Brood Mother: 18 HP. Sprite 40×40. Stationary. Spits homing projectiles (90 px/s, slight tracking, 1 dmg) every 2s in vulnerable phase, every 3s otherwise.

Nodes: 3 HP each. Stationary. Pulse visually when alive. Spawn 1 alien grunt per node every 6s while alive.

Adds: Alien grunts continuously while nodes are up. Spore drones occasionally drop from ceiling at random positions.

Act 3 Boss: Core Guardian

Arena: Cylindrical chamber surrounding the power core. Walls are organic and pulsing.

Phase 1 — Armored Guardian: 20 HP, but armored: only takes damage when 4 limb weak-points are exposed. Limbs glow yellow when armored, red when shootable. Each limb cycles armor on/off every 2s on staggered timers. Shoot a glowing-red limb 3 times to break it permanently. Once all 4 broken → phase 2.

Movement: Slow lateral pacing across the arena, 40 px/s. Stomps every 3s — telegraphed by sprite raise (0.5s warning), creates 24px shockwave AoE around feet for 1 dmg.

Phase 2 — Enraged Guardian: 15 HP, no armor. Speed 90 px/s. Two new attacks: (a) screen-wide horizontal sweep beam every 4s (telegraphed 1s in advance with red line), force player to move; (b) summons 2 stalkers at low HP (below 5).

Death: Drops the demolition charge (key item). Now the player has to plant it and run.

Act 4 "Boss" — The Pursuer

No traditional boss. Instead, in the final 2 screens of escape, a single Pursuer chases the player. The Pursuer is invulnerable. Sprite is a re-skinned Vessel (callback to act 1). Speed: matches player base speed. If it touches player → 1 dmg + i-frames. Pure tension; the player just has to outrun and reach the surface.

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8. ACT-BY-ACT CONTENT GUIDE

Act 1: Approach and Infiltration

Tone: Military thriller. Daylight to indoor fluorescent. Things feel wrong but human-explainable until the end.

Environment palette: Greys, dirt browns, military green. Concrete, chain-link, steel doors.

Screens:

1. **Outer perimeter** — chain link fence, body of a recon soldier holding a pistol (pickup), 2 grunts patrolling. *Tutorial: shoot, pick up.*
2. **Motor pool** — wrecked humvees as cover, 3 grunts, 1 sergeant. *Teach using cover.*
3. **Guard post entry** — narrow chokepoint, 2 grunts, locked door (clear room to open).
4. **Main hallway** — long corridor, 4 grunts, med-stim pickup.
5. **Mess hall** — overturned tables as cover, 2 sergeants and 2 grunts. *First "twitch" enemy here as easter egg corpse.*
6. **Barracks** — 3 grunts, 1 twitchy. *First introduction of twitchy enemy.* Story: scrawled writing on a bunk wall ("THEY KNOW. RUN.").
7. **Armory** — 2 sergeants, ammo pickups, grenade pickup (+1 max).
8. **Communications room** — 1 sergeant, 2 twitchy. Computer terminal flickers with alien text fragments.
9. **Command corridor** — 2 sergeants, 1 grunt. Final approach. Med-stim pickup before boss door.
10. **Command center** — Boss: General Hayes / The Vessel.

Act 2: Discovery

Tone: Things are wrong now. Concrete is being grown over by something. Fluorescent flickering, biolights pulsing.

Environment palette: Sickly greens, deep purples, infestation overlays on concrete.

Screens:

1. **Elevator descent** — short room, plasma rifle pickup, 2 hybrid soldiers as you exit elevator.
2. **Bunker corridor** — first half concrete, second half organic infestation. 3 hybrid soldiers.
3. **Old research lab** — lore screen. Dead scientist, computer log on screen ("BREACH SOURCE: SUBLEVEL 12. EVENT DATE: 1987."). 2 alien grunts.
4. **Infested barracks** — mirror image of act 1 r6 but covered in growths. 4 alien grunts.
5. **Hatchery hall** — 3 spore drones, 2 alien grunts. Hazardous.
6. **Pump room** — environmental: 2 spitters protect a passage. Walls have steam vents that periodically release damaging steam (telegraphed).
7. **Catwalk over chasm** — 4 hybrid soldiers, narrow walkway, fall damage if you step off (instant 1 HP loss + return to entry edge).
8. **Storage warehouse** — 2 sergeants (still some humans down here), 2 alien grunts, 4 spore drones. Most enemies in any act 2 room.
9. **Brood antechamber** — 3 alien grunts, plasma ammo pile, med-stim. Audible heartbeat sound builds.
10. **Brood chamber** — Boss: Brood Mother.

Act 3: The Descent

Tone: Pure alien. No human structures left. Bioluminescent glows in dark spaces.

Environment palette: Black backgrounds, glowing teals and magentas, organic curves.

Screens:

1. **Vertical shaft (entry)** — narrow, 2 stalkers, 1 spitter. The pace immediately ratchets up.
2. **Glowing cavern** — 3 stalkers, 2 drifters. Heavy laser pickup mid-room (rewarding exploration; tucked behind a wall).
3. **Spore corridor** — 4 spore drones, 2 spitters, narrow.
4. **Pulse chamber** — environmental: pulsing pillars do AoE damage on a 2s cycle (telegraphed, walk between pulses). 2 stalkers.
5. **Bridge over the abyss** — 3 drifters, 1 stalker. Drifters can pass through "porous" walls here, attack from off-platform.
6. **Hive corridor** — 2 spitters, 4 spore drones, 1 stalker. Final density spike.
7. **Hazard gauntlet** — no enemies, but moving lethal organisms (think conveyor of spike-tendrils) require timing. Med-stim and laser ammo.
8. **Core chamber** — Boss: Core Guardian. After kill, cinematic prompt "PRESS E TO PLANT CHARGE" → 5-second planting animation → escape act starts.

Act 4: Escape

Timer: 3:00 (180 seconds) on screen, prominent. Counts down. At 0:00, instant game over with "TOO LATE" and restart act 4.

Screen shake: Subtle constant shake intensifies as timer drops below 1:30, more aggressive below 0:30.

Visual treatment: Reuse act 3, 2, 1 rooms in reverse order, but with red emergency lighting overlay, debris particles falling, occasional flash from off-screen explosions.

Mechanics change: Wave spawners replace fixed enemies. When you enter a screen, 1-3 spawn points start emitting Swarmers at a rate of 1 every 1.5s. Optional act 2/3 enemies spawn occasionally. Killing all spawners (each is 2 HP) clears the screen and stops further waves. But you don't have to clear — you can run through.

Screens (escape route, reverse path):

1. Core chamber (now collapsing) — 2 spawners, swarmers, drifters.
2. Hive corridor (warped) — 3 spawners, swarmers + spore drones.
3. Glowing cavern (warped) — 1 spawner + 2 stalkers loose.
4. (Back into act 2 territory) Brood chamber (now empty, just rubble) — fast traverse.
5. Catwalk (sections collapsing) — gaps open up; jump-equivalents (sprint over).
6. Bunker corridor — 2 spawners + hybrid soldiers.
7. (Surface) Main hallway — 3 spawners, dense waves of swarmers + the single PURSUER appears here.
8. Outer perimeter — final sprint, Pursuer is right behind you, reach the helicopter LZ marker = win.

End screen: "EXTRACTION SUCCESSFUL." Time taken, deaths, secrets found stats. Press R to restart.

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9. VISUAL STYLE GUIDE

Palette (16 colors, NES-inspired):

- 0: `#000000` black (background, outlines)
- 1: `#1a1a2e` deep navy (act 3 bg)
- 2: `#16213e` dark blue (act 2 bg)
- 3: `#3a2e1a` dark brown (act 1 bg accents)
- 4: `#7a7a7a` grey concrete
- 5: `#c8c8c8` light grey (concrete highlights)
- 6: `#5a3a1a` brown (military fatigues)
- 7: `#8b6914` khaki (uniforms)
- 8: `#2d5016` dark green (military)
- 9: `#7fb069` glow green (alien bioluminescence)
- 10: `#d62828` blood red (damage flash, alarms)
- 11: `#f77f00` orange (explosions, warnings)
- 12: `#fcbf49` yellow (laser, telegraphs)
- 13: `#7209b7` purple (alien organic)
- 14: `#f72585` magenta (alien glow)
- 15: `#ffffff` white (highlights, UI text)

Per-act background tinting: Apply a subtle color overlay to the entire frame to shift the mood. Act 1: neutral. Act 2: green/sickly. Act 3: deep purple/teal. Act 4: red emergency.

Sprite style: Chunky, 1-2 colors per sprite plus an outline. Player has a visible direction-facing indicator (gun barrel pointing the aim direction). Enemies have a clear silhouette readable at small size.

HUD:

- Top-left: 5 health pips (red squares, dim when lost)
- Top-right: current weapon name + ammo count
- Bottom-left: grenade count
- Top-center (act 4 only): timer in red
- Bottom-right: stamina bar (only visible while sprinting or recovering)

Particles: Hit feedback = 4-6 small white sparks for 0.2s. Death explosion = 12-16 colored particles for 0.5s. Bullet impact on wall = 2-3 dust particles for 0.15s.

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10. AUDIO GUIDE (PROCEDURAL)

All Web Audio. Build a small `Sfx` module with these sound functions:

Sound	Recipe
Pulse rifle shot	Square wave, 880 Hz → 220 Hz sweep over 0.06s, gain envelope 0.3 → 0
Pistol shot	Square, 1100 Hz → 400 Hz over 0.04s
Plasma shot	Sawtooth, 220 Hz held 0.12s, with low-pass filter sweeping down
Laser	Sine 880 Hz + saw 220 Hz, 0.3s, ring-mod feel
Grenade explosion	White noise burst, low-pass 800 Hz, 0.4s envelope
Enemy hit	Noise burst 0.05s, high-pass 2000 Hz
Enemy death	Noise burst 0.2s, low-pass sweeping down from 1500
Player hit	Square 110 Hz, 0.15s, with detune wobble
Pickup	Sine 660 → 990 Hz blip
Door unlock	Two-note rising blip
Boss phase transition	Low rumble (sine 60 Hz) + reverse cymbal (filtered noise rising)

Per-act ambient drone (instead of music):

- Act 1: Sine 110 Hz + filtered noise (radio static feel), low volume
- Act 2: Detuned dual sine 90/93 Hz + occasional high pings (drips)
- Act 3: Sine 60 Hz + slow LFO on cutoff filter (alive/breathing feel)
- Act 4: Pulsing siren (sine 440↔660 Hz square LFO) + low rumble

Volume should be configurable; default ambient at 30% of SFX volume.

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11. PROGRESSION PACING

Time	Player should have
0:00	Pulse rifle, 1 grenade, 5 HP
1:30 (mid act 1)	+ Pistol
4:00 (start act 2)	+ Plasma rifle, ~30 plasma ammo
8:00 (start act 3)	All weapons. Heavy laser by mid-act 3.
12:00 (start act 4)	Whatever they have. Starting ammo refilled.
15-18:00	Win.

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12. CODE ARCHITECTURE — RECOMMENDED FILE STRUCTURE

Even though it's a single HTML file, organize the JS into these logical sections (in order, separated by clear `// =====` comment banners):

1. **Constants** — palette, dimensions, gameplay numbers
2. **Sprite data** — all sprite arrays
3. **Room data** — all 36 rooms in a single object/array
4. **Audio system** — Web Audio init, Sfx functions, ambient
5. **Input system** — keyboard listeners, input state
6. **Entity classes** — Player, Bullet, Enemy (with subclasses), Boss, Pickup, Particle, Spawner
7. **Room manager** — load, transition, clear-check
8. **Game loop** — update, render, fixed timestep (60fps)
9. **UI/HUD** — health, ammo, timer, menus, death screen, win screen
10. **Bootstrap** — init, attach to canvas, start

Game loop: Use `requestAnimationFrame` with a fixed-timestep accumulator (1/60s). Decouple update from render so the game runs the same speed on any monitor.

State machine: Top-level states: `MENU`, `PLAYING`, `PAUSED`, `DYING`, `ACT_TRANSITION`, `WIN`, `LOSE`. Clean switch in the main loop.

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13. BUILD ORDER FOR CLAUDE CODE

Don't try to build everything at once. Suggested order:

1. **Skeleton.** HTML file, canvas, game loop, palette constant, render a single static room (4 walls, player rectangle in middle).
2. **Player movement.** WASD, collision with walls. No shooting yet.
3. **Player shooting.** Arrow keys, bullets that travel and despawn off-screen.
4. **One enemy.** Grunt — patrols, shoots at player, dies in 2 hits, damages player on hit. I-frames working.
5. **Room transitions.** Build 3 connected rooms, walk between them, exits/walls work.
6. **Locked exits.** Combat rooms lock until enemies dead.
7. **HUD.** Health, ammo, weapon display.
8. **Audio system.** Get the shot/hit/death sounds in.
9. **Build out act 1.** All 10 rooms, all 3 act-1 enemy types. Boss fight (both phases).
10. **Pickups, weapon switching, grenades.** Tab to cycle, space to throw.
11. **Build out act 2.** New enemies, brood mother boss.
12. **Build out act 3.** New enemies, environmental hazards, core guardian.
13. **Act 4.** Wave spawners, timer, escape sequence, pursuer, end screen.
14. **Polish pass.** Particles, screen shake, ambient drones, transitions, death/restart flow, menu screen, win screen.

Each step should produce a runnable game, even if incomplete. Test before moving on.

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14. PLAYTEST TARGETS

When the build is functional, validate against these:

- A first-time player should reach end of act 1 in under 5 minutes without dying more than twice.
- Twin-stick controls should feel natural within 30 seconds.
- Each act's enemies should feel distinct (not just reskins).
- The act 1 boss transformation should genuinely surprise.
- The brood mother fight should require killing nodes — players who try to just shoot her should learn the mechanic in under 30 seconds.
- The escape timer should feel desperate but reach-able.
- Total first playthrough: 15-25 minutes for a competent player.

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15. EXPLICITLY OUT OF SCOPE FOR V1

Do not build:

- Save system / persistence
- Difficulty options
- Mouse/gamepad input
- Music (drones only)
- Multiplayer
- Level editor
- Mobile/touch controls
- Externally loaded assets
- Cinematics or cutscenes (text dialogue only)
- Unlockables / progression between runs

If any of these come up, defer to v2.

END OF SPEC

Build this in a single `index.html` file. When in doubt about a number, defaults are deliberately tuned in this doc — pick the value listed, ship it, tune later if a playtest demands it. Prioritize making the whole thing playable end-to-end before polishing any individual part.